ConstructionSafety-FaithBench: Auditing Visual-Evidence Faithfulness in Construction-Safety Vision-Language Models

Abstract

Construction-safety vision-language models (VLMs) are being explored as tools for reviewing site imagery, but standard evaluation usually asks only whether the final answer or rationale sounds correct. This leaves a critical gap: a model may predict a safety violation while grounding on irrelevant scene context, or keep the same answer after the rule-relevant worker, protective equipment, or hazard region is removed. We address this gap with ConstructionSafety-FaithBench, a reproducible audit pipeline that separates answer correctness, visual evidence localization, and counterfactual sensitivity. The artifact contains 163 pilot image-rule pairs, 588 scale-up candidates, four safety-rule families, deterministic baselines, a Florence-2 grounding adapter, 948 targeted and matched-random occlusion runs, and a 120-row final audit-label layer. The audit set prioritizes Florence/bootstrap disagreements and weak evidence overlap. Against final audit labels, Florence grounding reaches 18.3% accuracy and 12.5% macro-F1, while a metadata-assisted bootstrap reaches 78.3% accuracy, showing that scaffold-level labels can mask visually grounded failure. Targeted evidence occlusion changes Florence answers 39.2% of the time versus 9.2% for matched random masks, a paired gap of 30.0 percentage points. The final audit labels combine 108 role-conditioned A/B consensus rows with 12 returned adjudication decisions, so results are reported with explicit provenance rather than as unqualified human ground truth.

Keywords: trustworthy AI; explainable AI; vision-language models; construction safety; evidence faithfulness; AI risk assessment

1. Introduction

Construction safety decisions depend on small, localized, rule-specific visual cues: a hard hat on a worker, a guardrail at an exposed edge, a harness at height, or equipment near a pedestrian path. Vision-language models are attractive for screening such imagery because they can combine natural-language rules with visual inputs. However, a correct final answer is not sufficient for safety-facing use. A model can identify a hard-hat violation while grounding its response on background scaffolding, or it can preserve the same answer after the relevant worker or protective equipment is masked.

This creates an evaluation gap. Most VLM evaluation reports answer accuracy, generated rationales, or object boxes as separate signals, but construction-safety review needs to know whether these signals agree. If the answer is correct but the evidence is irrelevant, the model may be relying on scene priors. If the answer is stable when rule-relevant evidence is removed, the model may be insensitive to the visual cue it claims to use. Standard answer-only scoring merges these failure modes and can overstate the reliability of a safety pipeline.

We address this gap with ConstructionSafety-FaithBench, a reproducible audit pipeline for visual-evidence faithfulness in construction-safety VLMs. Each example pairs a construction-site image with a natural-language safety rule, answer label, evidence regions, and provenance fields. The benchmark focuses on four rule families: hard-hat/PPE compliance, harness/fall protection, guardrail/edge protection, and struck-by/equipment proximity. The pipeline produces normalized manifests, model outputs, audit labels, intervention specifications, score tables, figures, and release manifests.

The study evaluates deterministic baselines and a Florence-2 grounding adapter, then tests whether targeted rule-evidence masks affect model behavior more than same-size matched-random masks. The results show why evidence-level auditing matters: Florence performs poorly on the prioritized audit set, and targeted occlusion produces substantially larger answer changes than random masking. The paper contributes a compact benchmark structure, a final audit-label layer with explicit provenance, and an intervention analysis that separates answer correctness from evidence dependence.

2. Benchmark and Audit Pipeline

Each row contains an image identifier, source split, natural-language safety question, normalized rule identifier, answer label, evidence boxes, and provenance metadata. Raw images and model weights are not redistributed in the repository; reproducible scripts load source images only when

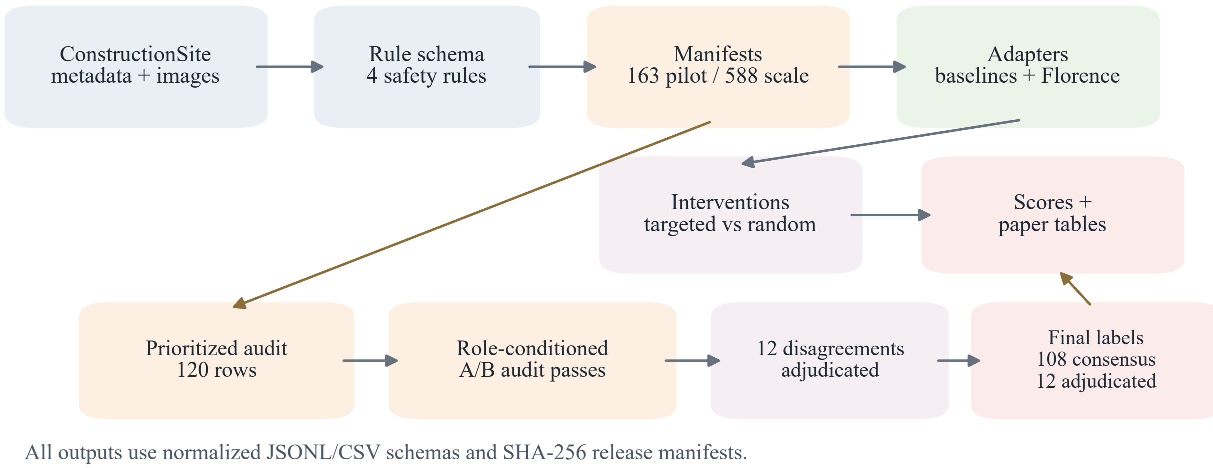

Figure 1: Benchmark architecture. The same schema supports manifests, model outputs, audit labels, interventions, score tables, and release manifests.

local dataset access is available.

Figure 1 summarizes the architecture. ConstructionSite-derived metadata and images are converted into pilot and scale-up manifests. Model adapters write normalized CSV/JSONL outputs. Scores are computed against either model-assisted bootstrap labels or the completed final audit-label layer. Intervention specifications reuse evidence boxes to compare targeted rule-evidence masks with same-size matched-random masks.

The annotation stack has three layers. First, model-assisted bootstrap annotations use source metadata, captions, rule-violation records, and available boxes. These labels support scaling and triage, but they are not ground truth. Second, a balanced 120-row audit queue samples 30 rows per rule from difficult scale-up cases: 118 of 120 rows are Florence/bootstrap disagreements. Third, the final audit-label layer resolves the queue. It contains 108 rows where two independent role-conditioned audit passes agreed and 12 rows resolved by returned adjudication. Final labels are 83 violation, 31 compliant, and 6 uncertain.

3. Metrics and Experimental Setup

The benchmark reports three metric families. Answer metrics include accuracy and macro-F1 across compliant, violation, and uncertain labels. Evidence metrics include evidence presence, best IoU, normalized centroid drift, and evidence disappearance. Intervention metrics compare a targeted mask over rule-relevant evidence against five same-size random masks sampled per image. The paired statistic is the targeted effect minus the within-image matched-random mean.

The implemented adapters are intentionally simple and auditable. The annotation bootstrap echoes the metadata-assisted label. Manifest seed and majority-violation baselines test weak-label and class-prior behavior. A caption-keyword baseline checks whether captions alone leak the labels. The Florence grounding adapter runs *microsoft/Florence-2-base-ft* and maps grounded detections through deterministic geometry. Each adapter records normalized outputs and provenance.

The intervention study contains 948 Florence reruns: 158 targeted occlusions and 790 matched-random controls. Random masks use the same clipped width and height as the target mask, preserving mask area while changing location. Bootstrap intervals and paired tests are computed at the image level.

4. Results

Figure 2 shows the completed audit layer and answer scores. The final-label layer exposes a sharp difference between scaffold validation and visually grounded audit behavior. The metadata-assisted bootstrap reaches 78.3% accuracy against final labels, while Florence grounding reaches only 18.3% accuracy and 12.5% macro-F1. The two role-conditioned passes are high against the final layer because most final labels are consensus rows and the remaining disagreement rows were adjudicated.

Table 1: Headline results from the current audit and intervention study.

Quantity	Value
Final audit rows	120
Final audit label counts	83 V / 31 C / 6 U
A/B agreement rows	108
Returned adjudication rows	12
Florence accuracy on final audit labels	18.3%
Bootstrap accuracy on final audit labels	78.3%
Targeted answer-flip rate	39.2%
Matched-random answer-flip rate	9.2%
Paired answer-flip difference	30.0 points
Paired centroid-drift difference	0.160

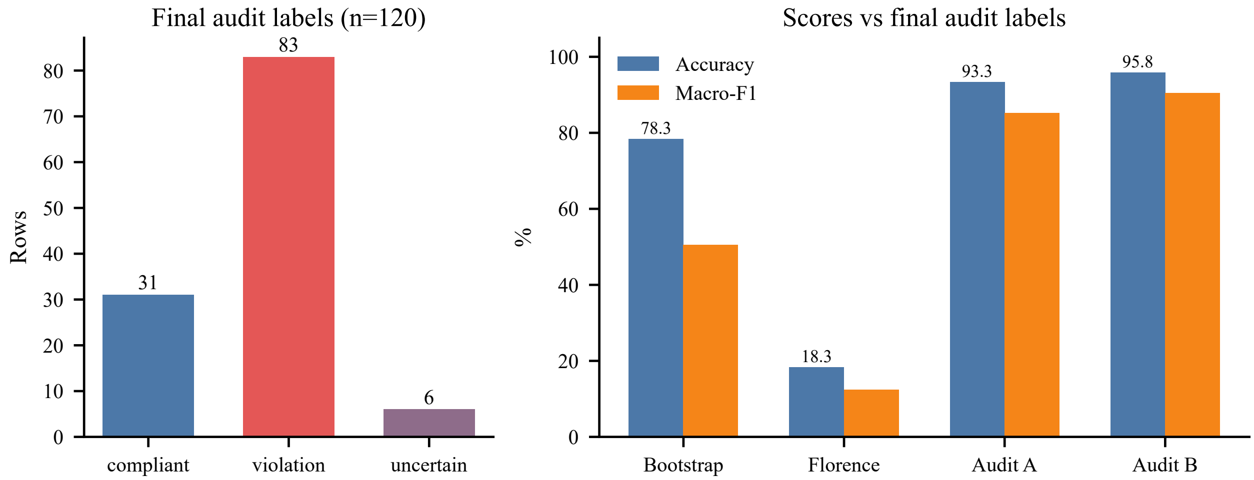

Figure 2: Final audit labels and answer scores. The audit split intentionally over-samples difficult disagreement and weak-overlap cases.

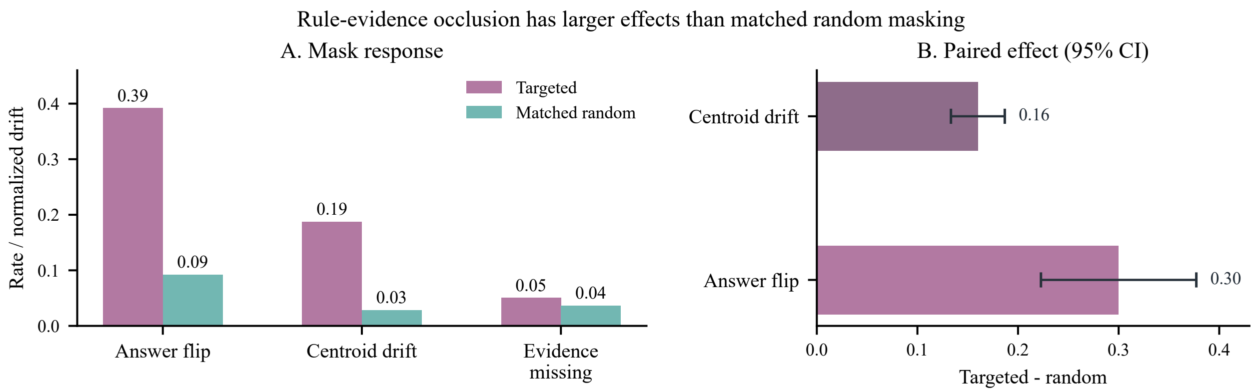

Figure 3: Targeted rule-evidence masking affects answers and evidence more than same-size random masking.

Figure 3 summarizes the counterfactual faithfulness result. Targeted rule-evidence occlusion flips Florence answers 39.2% of the time, compared with 9.2% for same-size random masks. The paired answer-flip gap is 30.0 percentage points with a bootstrap 95% interval of 22.3 to 37.7 points. Targeted masks also produce larger evidence relocation: the paired centroid-drift difference is 0.160 with a 95% interval of 0.133 to 0.187. Evidence disappearance is tracked separately because a missing post-mask box cannot be treated as ordinary IoU drift.

The scale-up results add a diagnostic pattern. Florence often returns some evidence region, yet answer accuracy remains weak and rule-dependent. This means evidence presence alone is not enough; a model can emit boxes without using the right safety cue or producing the right answer. The strongest paper-level finding is therefore not that Florence is reliable, but that the benchmark separates failure modes that ordinary answer scoring would merge.

5. Discussion and Implications

The results support a compact but useful measurement claim. Answer correctness, visual evidence, and counterfactual sensitivity should be reported separately for construction-safety VLMs. The final audit set shows that a model can look superficially functional under bootstrap labels yet fail badly on cases selected for disagreement and weak evidence overlap. The intervention study shows that targeted evidence masks have larger effects than same-size random masks, suggesting location-specific dependence but not proving recovery of internal reasoning.

For trustworthy innovation, the practical implication is governance rather than automation. A safety-support interface should show the cited region, state when evidence is missing or unstable, preserve uncertainty, and keep final authority with qualified safety personnel. The benchmark can support model comparison, regression testing, and audit documentation, but it cannot justify worker discipline, surveillance, or autonomous inspection.

6. Limitations and Ethics

The current study is not a deployment validation. The final audit labels are complete, but the A/B passes are role-conditioned audit passes and the adjudication package must be reported with that provenance. Future work should replace or verify this layer with independent domain-expert human annotation. The model set is also limited: Florence-2 is the only image-grounded adapter, and stronger instruction-following VLMs should be added before broad model-ranking claims. The scale-up split remains imbalanced because struck-by and fall-hazard examples are scarce.

Construction imagery raises privacy, surveillance, and labor-governance concerns. The benchmark does not infer identity, intent, competence, or blame. It should not be used for autonomous inspection or disciplinary monitoring. Appropriate use is research evaluation under documented scope limits, data minimization, source-license compliance, and human professional review.

7. Conclusion

ConstructionSafety-FaithBench turns construction-safety VLM evaluation from answer checking into an evidence-faithfulness audit. The current results show that Florence grounding performs poorly on a difficult final-label audit set and that targeted evidence masks affect answers more than matched random masks. The benchmark is therefore useful as a compact, reproducible testbed for trustworthy visual-evidence evaluation, while remaining explicitly outside autonomous safety deployment.

References

Chen, X., & Zou, Z. (2025). Are large pre-trained vision language models effective construction safety inspectors? *arXiv*. https://arxiv.org/abs/2508.11011

Howard, J., & Schulte, P. A. (2024). Exploring approaches to keep an AI-enabled workplace safe for workers. *National Institute for Occupational Safety and Health*. https://www.cdc.gov/niosh/bulletin/2024/ai-risk-management.html

Jain, S., & Wallace, B. C. (2019). Attention is not explanation. In *Proceedings of NAACL-HLT 2019* (pp. 3543–3556). https://doi.org/10.18653/v1/N19-1357

Li, K., Vosselman, G., & Yang, M. Y. (2025). Multimodal rationales for explainable visual question answering. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshops* (pp. 191–201).

National Institute of Standards and Technology. (2023). *Artificial Intelligence Risk Management Framework (AI RMF 1.0)*. https://doi.org/10.6028/NIST.AI.100-1

Phillips, P. J., Hahn, C. A., Fontana, P. C., Yates, A. N., Greene, K., Broniatowski, D. A., & Przybocki, M. A. (2021). *Four principles of explainable artificial intelligence* (NISTIR 8312). National Institute of Standards and Technology. https://doi.org/10.6028/NIST.IR.8312

Wu, J., Kang, W., Tang, H., Hong, Y., & Yan, Y. (2024). On the faithfulness of Vision Transformer explanations. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition* (pp. 10936–10945).

Xiao, B., Wu, H., Xu, W., Dai, X., Hu, H., Lu, Y., Zeng, M., Liu, C., & Yuan, L. (2024). Florence-2: Advancing a unified representation for a variety of vision tasks. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition* (pp. 4818–4829).